CSE 4111
Introduction to Data Science
Chapter 1: Introduction

COURSE COORDINATOR

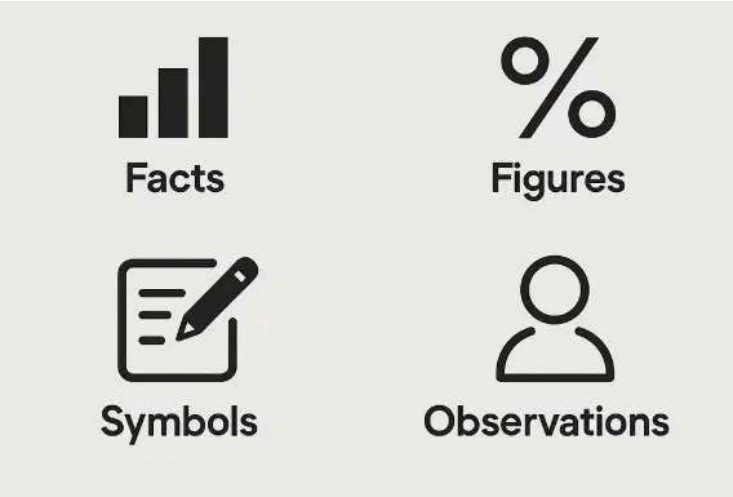
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Outlines

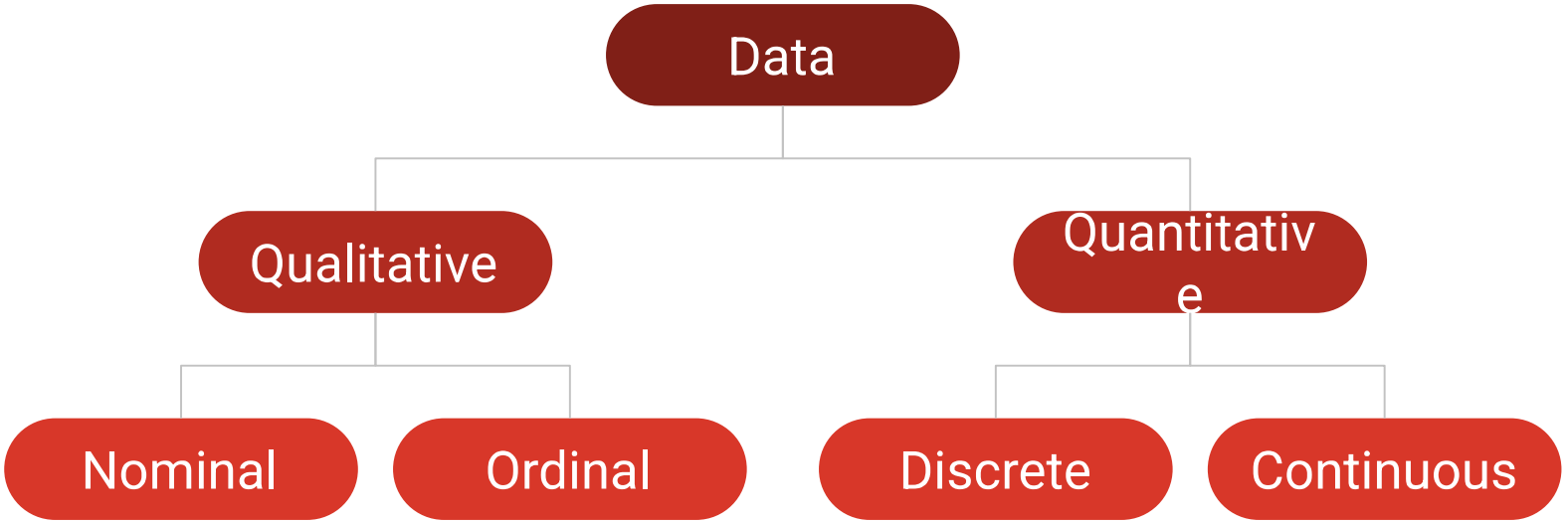
- ❑ Data
- ❑ Big Data
- ❑ Data Science
- ❑ Application of Data Science
- ❑ Data Science Process Life Cycle
- ❑ Role of a Data Scientist

Data

Data is a collection of raw, unprocessed facts, figures, symbols, or observations such as numbers, words, or images that represent information about events, objects, or phenomena



Types of data



Qualitative or Categorical Data

- Describes data in categories (non-numerical)
- Based on labels or names, not numbers
- Examples: gender, hometown, favourite sport

Key Idea

- Sometimes includes numbers without mathematical meaning
  (they act as labels, not quantities)

Examples

- Birthdate, school postcode (numbers used as identifiers, not for calculation)

Qualitative or Categorical Data

Nominal Data:

- Used to label variables (no numerical meaning)
- Also called Nominal Scale
- Cannot be ordered or measured
- Examples: gender, letters, symbols, words

Analysis & Representation

- Analyzed by grouping into categories
- Measured using frequency or percentage
- Visualized using pie charts

Qualitative or Categorical Data

Ordinal Data:

- Data with a natural order or ranking
- Difference between values is not measurable
- Common in surveys, questionnaires, finance, economics

Representation & Analysis

- Visualized using bar charts
- Also shown in tables (categories in rows)
- Interpreted using various visualization tools

Example:

Economic status (“low income”, ”middle income”, ”high income”)

Education level (“high school”,”BS”,”MS”, ”PhD”)

Income level (“less than 50K”, “50K-100K”, “over 100K”)

Satisfaction rating (“extremely dislike”, “dislike”, “neutral”, “like”, “extremely like”).

Quantitative(Neumerical) data

- Represents numbers (how much, how many, how often)
- Shows quantity or measurement
- Examples: height, weight, length, size

Types of Quantitative Data

1. Discrete Data

- Takes countable, finite values
- Cannot be divided further
- Example: number of students

2. Continuous Data

- Takes infinite values within a range
- Can be measured precisely
- Example: temperature range

Big Data

- Refers to massive, fast-growing, and complex data
- Too large for traditional processing tools

Types of Big Data

- Structured → tables, databases
- Semi-structured → JSON, XML
- Unstructured → text, images, videos

Why It Matters

- Data is growing rapidly due to devices, sensors, online platforms
- Requires advanced tools & technologies for processing and analysis

Characteristics of Big Data

The 4 V's of Big Data

- **Volume**: Refers to the huge amount of data generated every second-ranging from terabytes to petabytes. Example: YouTube uploads 500+ hours of video every minute.
- **Velocity**: The speed at which data is created, shared, and processed. Data streams in from sensors, social media, and transactions in real-time.
- **Variety**: Data comes in multiple formats-text, audio, images, videos, logs, sensor data, etc. Handling all these types together is complex
- **Veracity**: Refers to the trustworthiness and accuracy of the data. Inconsistent, duplicated, or noisy data can lead to wrong insights.

Characteristics of Big Data

Additional V's:

- **Value**: Not all data is useful. The key is extracting relevant data and turning it into business value through analytics.
- **Variability**: Data meaning may change over time or context.
- **Visualization**: Making complex data understandable through visual tools (charts, graphs, dashboards).

Hadoop

- **Open-source framework** for storing & processing large data
- Works in a **distributed environment** (many machines)
- Based on **MapReduce** → enables parallel processing

Key Features

- **Scalable** → can handle growing data easily
- **Fault-tolerant** → continues even if some nodes fail
- **Cost-effective** → uses inexpensive machines (clusters)

Technology

- Built mainly using **Java** (with C & shell scripts)

Hadoop Architecture

Hadoop has two main components:

- **Hadoop Distributed File System (HDFS):** HDFS breaks big files into blocks and spreads them across a cluster of machines. This ensures data is replicated, fault-tolerant and easily accessible even if some machines fail.
- **MapReduce:** MapReduce is the computing engine that processes data in a distributed manner. It splits large tasks into smaller chunks (map) and then merges the results (reduce), allowing Hadoop to quickly process massive datasets.

Working Procedure of Hadoop

1. *Data is loaded into HDFS, where it's split into blocks and distributed across DataNodes.*
2. *MapReduce jobs are submitted to the ResourceManager.*
3. *The job is divided into map tasks, each working on a block of data.*
4. *Map tasks produce intermediate results, which are shuffled and sorted.*
5. *Reduce tasks aggregate results and generate final output.*
6. *The results are stored back in HDFS or passed to other applications.*

Data Science

Data Science is a combination of multiple disciplines that uses statistics, data analysis, and machine learning to analyze data and to extract knowledge and insights from it.
Data Science is about data gathering, analysis and decision-making.

Data Science is about finding patterns in data, through analysis, and make future predictions.

By using Data Science, companies are able to make:

- Better decisions (should we choose A or B)
- Predictive analysis (what will happen next?)
- Pattern discoveries (find pattern, or maybe hidden information in the data)

Application of Data Science

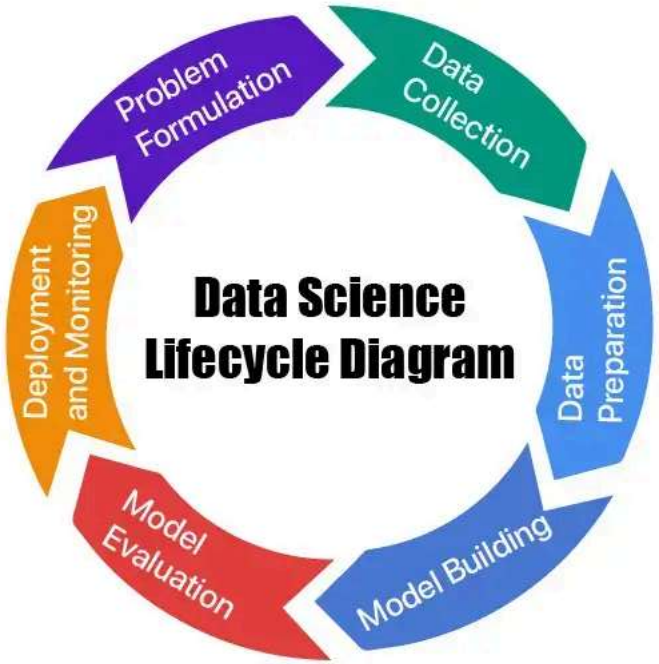
- For route planning: To discover the best routes to ship
- To foresee delays for flight/ship/train etc. (through predictive analysis)
- To create promotional offers
- To find the best suited time to deliver goods
- To forecast the next years revenue for a company
- To analyze health benefit of training
- To predict who will win elections

Data Science can be applied in nearly every part of a business where data is available. Examples are:

- Consumer goods
- Stock markets
- Industry
- Politics
- Logistic companies
- E-commerce

And many more…

Data Science Process Life Cycle



Data Science Process Life Cycle

1. Problem Formulation

This step defines the problem clearly and identifies the goal of the project.

Example

A hospital wants to predict whether a patient has diabetes based on medical data.

Key Tasks

- Understand business or research goal
- Define input and output
- Decide success criteria

Data Science Process Life Cycle

In this step, relevant data is gathered from different sources.

Example

Collect patient data such as:

- Age
- Blood sugar
- BMI
- Blood pressure

Data Sources

- Databases
- APIs
- Surveys
- Sensors
- CSV files

Data Science Process Life Cycle

3. Data Preparation

Raw data is cleaned and transformed into usable format.

Example

- Remove missing values
- Handle outliers
- Normalize data
- Convert categorical data into numeric form

Example Task

If salary values are very large compared to age values, normalization may be applied.

Data Science Process Life Cycle

4. Model Building

A machine learning or statistical model is trained using prepared data.

Example

Use:

- Logistic Regression
- Decision Tree
- Random Forest

to predict diabetes.

Key Tasks

- Split data into training and testing sets
- Train the model
- Tune parameters

Data Science Process Life Cycle

5. Model Evaluation

The model's performance is tested using evaluation metrics.

Example

Evaluate diabetes prediction model using:

- Accuracy
- Precision
- Recall
- F1-score

If model accuracy is 92%, it means the model predicts correctly 92 out of 100 cases.

Data Science Process Life Cycle

6. Deployment

The trained model is integrated into a real-world system.

Example

A hospital web application uses the model to predict diabetes when doctors enter patient information.

Deployment Platforms

- Web apps
- Mobile apps
- Cloud services

Role of a Data Scientist

- **Data Collection and Preparation**: Gather and clean raw data to make it ready for analysis
- **Pattern Identification**: Use statistics and algorithms to find trends and meaningful patterns
- **Tools and Technologies:** Work with tools like Python, NumPy, Pandas and Matplotlib for analysis
- **Predictive Modeling:** Build models to forecast future outcomes and support planning
- **Data Visualization:** Create charts and reports to present insights clearly
- **Data Accuracy:** Ensure data quality and consistency to avoid errors
- **Stakeholder Communication:** Present insights in a simple way to support decision making

Thank you

for your Attention